COMPASS_TEMPEST_SGW_2023: June & TMP III

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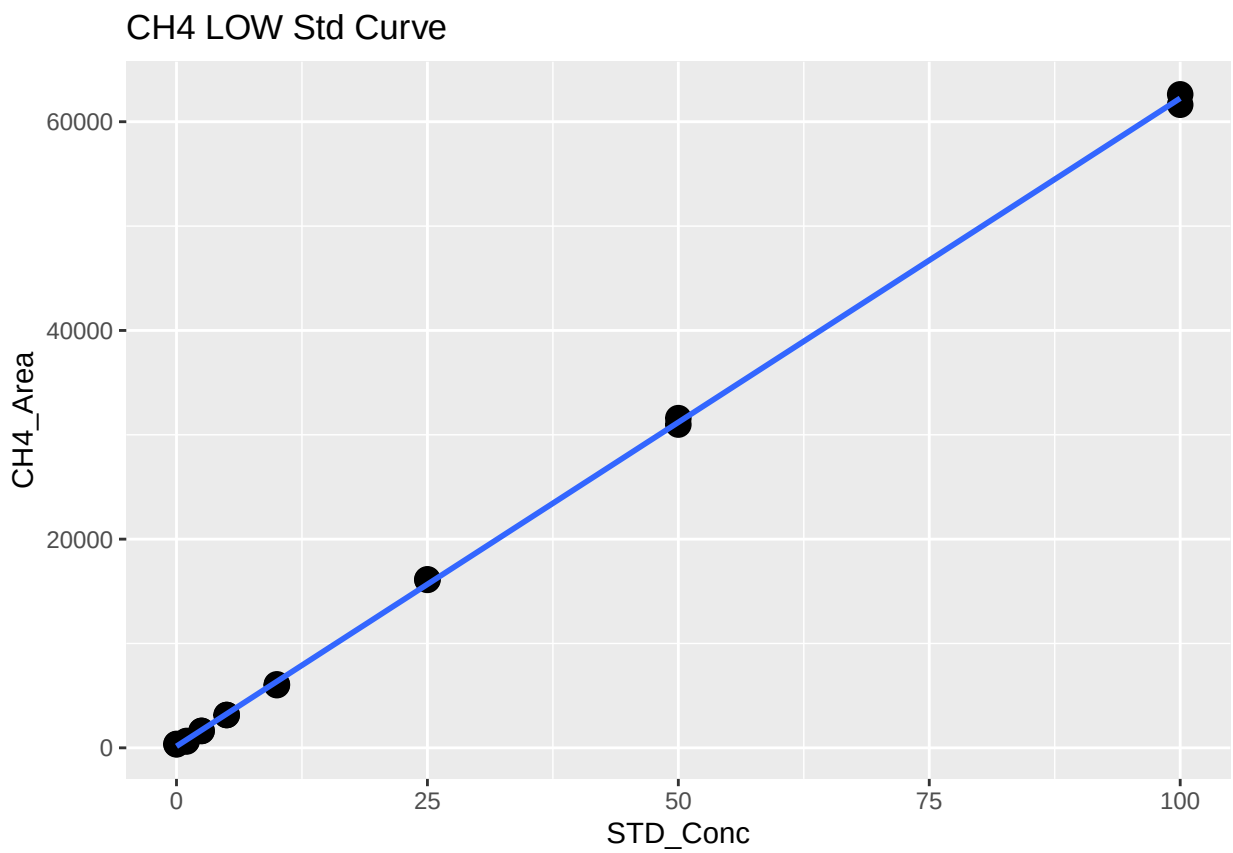
##Set Up

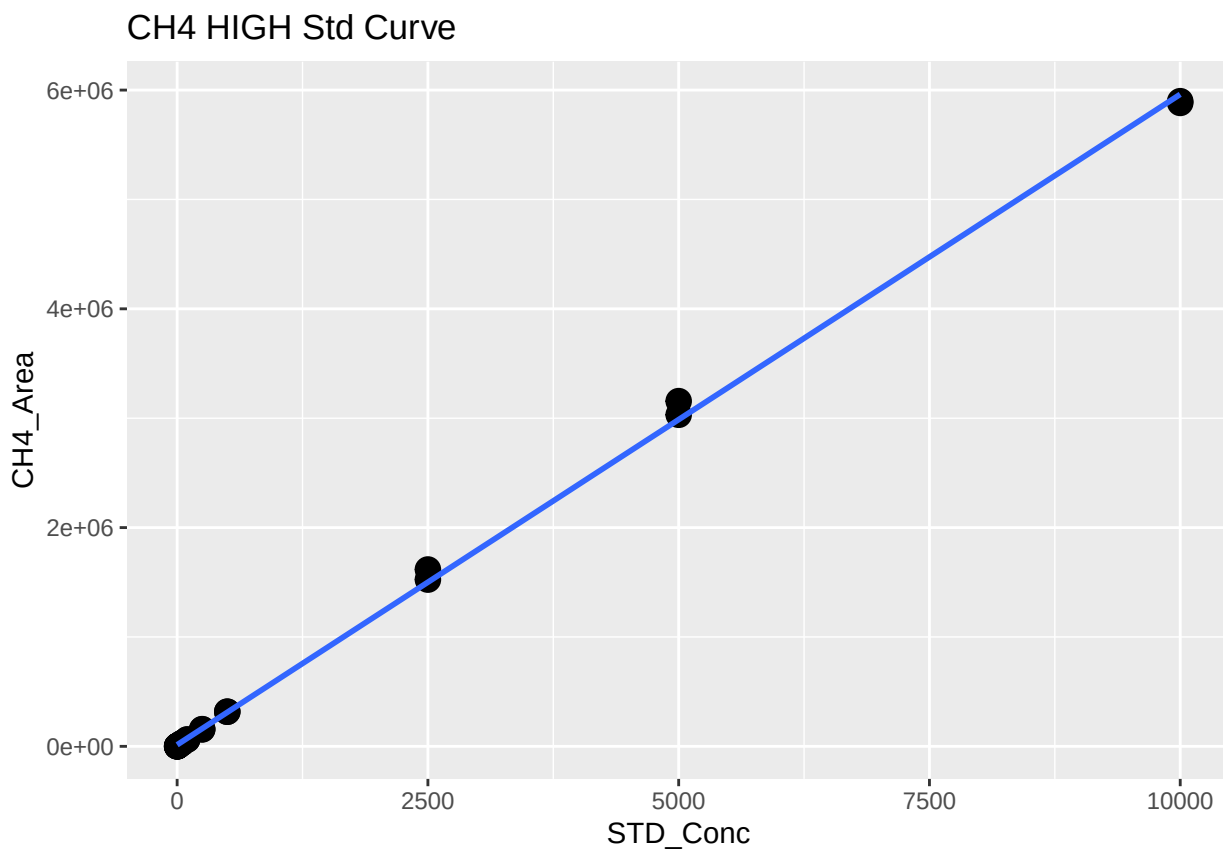
Read in first data file and assess standard curves

```
##     Machine       User Run_Date Sample_Type Sample1 Sample_Year Sample_Month
## 1 Varian GC A. Stearns 20240722       Blank   Blank        2024         <NA>
## 2 Varian GC A. Stearns 20240722     STD_CH4 STD_CH4        2024         <NA>
## 3 Varian GC A. Stearns 20240722     STD_CO2 STD_CO2        2024         <NA>
## 4 Varian GC A. Stearns 20240722     STD_CH4 STD_CH4        2024         <NA>
## 5 Varian GC A. Stearns 20240722     STD_CH4 STD_CH4        2024         <NA>
## 6 Varian GC A. Stearns 20240722     STD_CH4 STD_CH4        2024         <NA>
##   Sample_Day Event_Stamp Sample_Time Time_Zone G_W Sample_Plot Sample_Grid
## 1         NA        <NA>          NA      <NA>   G        <NA>        <NA>
## 2         NA        <NA>          NA      <NA>   G        <NA>        <NA>
## 3         NA        <NA>          NA      <NA>   G        <NA>        <NA>
## 4         NA        <NA>          NA      <NA>   G        <NA>        <NA>
## 5         NA        <NA>          NA      <NA>   G        <NA>        <NA>
## 6         NA        <NA>          NA      <NA>   G        <NA>        <NA>
##   Sample_Depth            Sample_ID Dilution_Factor STD_Conc CO2_Area CH4_Area
## 1         <NA>              Blank_0               1      0.0        0        0
## 2         <NA> Blank_0_repeatforCH4               1      0.0        0        0
## 3         <NA> Blank_0_repeatforCO2               1      0.0        0        0
## 4         <NA>         STD_1ppm_CH4               1      1.0    21489      457
## 5         <NA>       STD_2.5ppm_CH4               1      2.5   144771     1168
## 6         <NA>         STD_5ppm_CH4               1      5.0   331023     4703
##   Field_Notes Lab_Notes
## 1          NA
## 2          NA
## 3          NA
## 4          NA
## 5          NA
## 6          NA
```

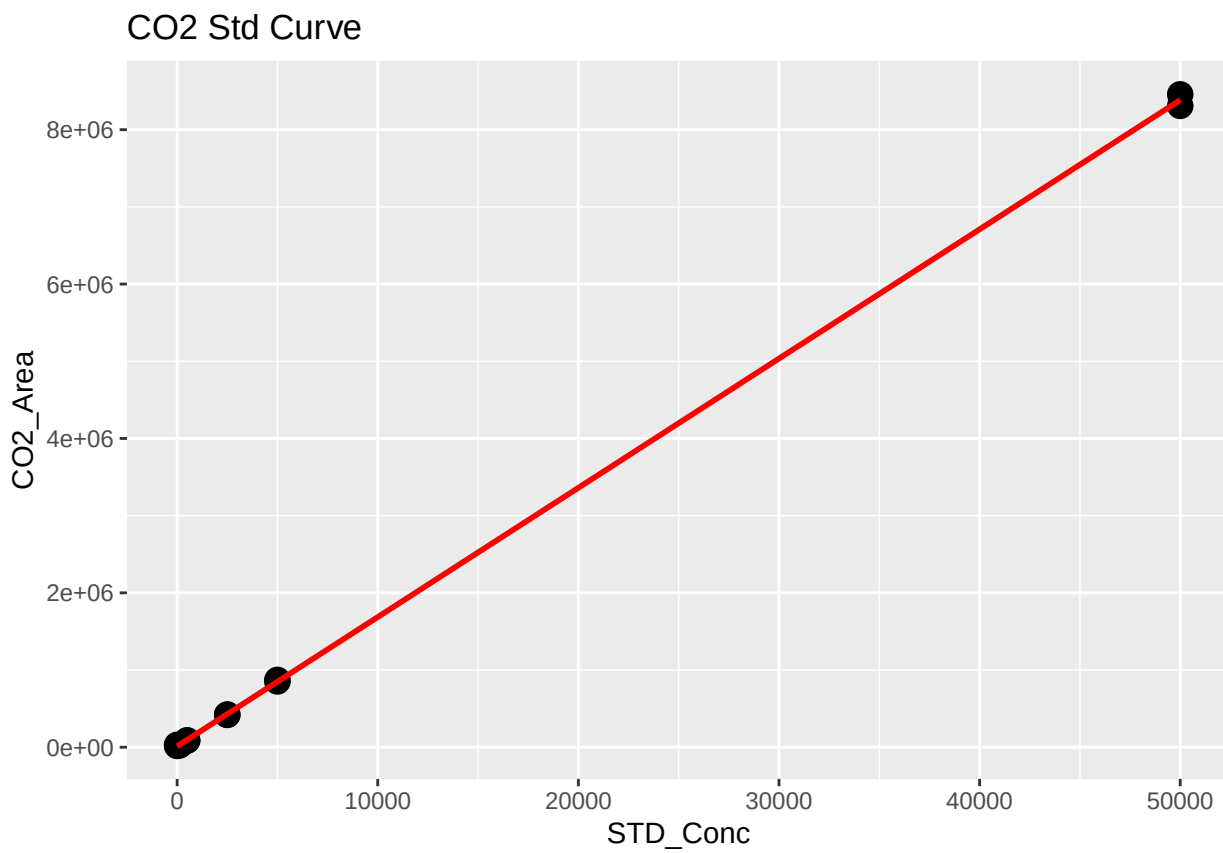
```
## `geom_smooth()` using formula = 'y ~ x'
```



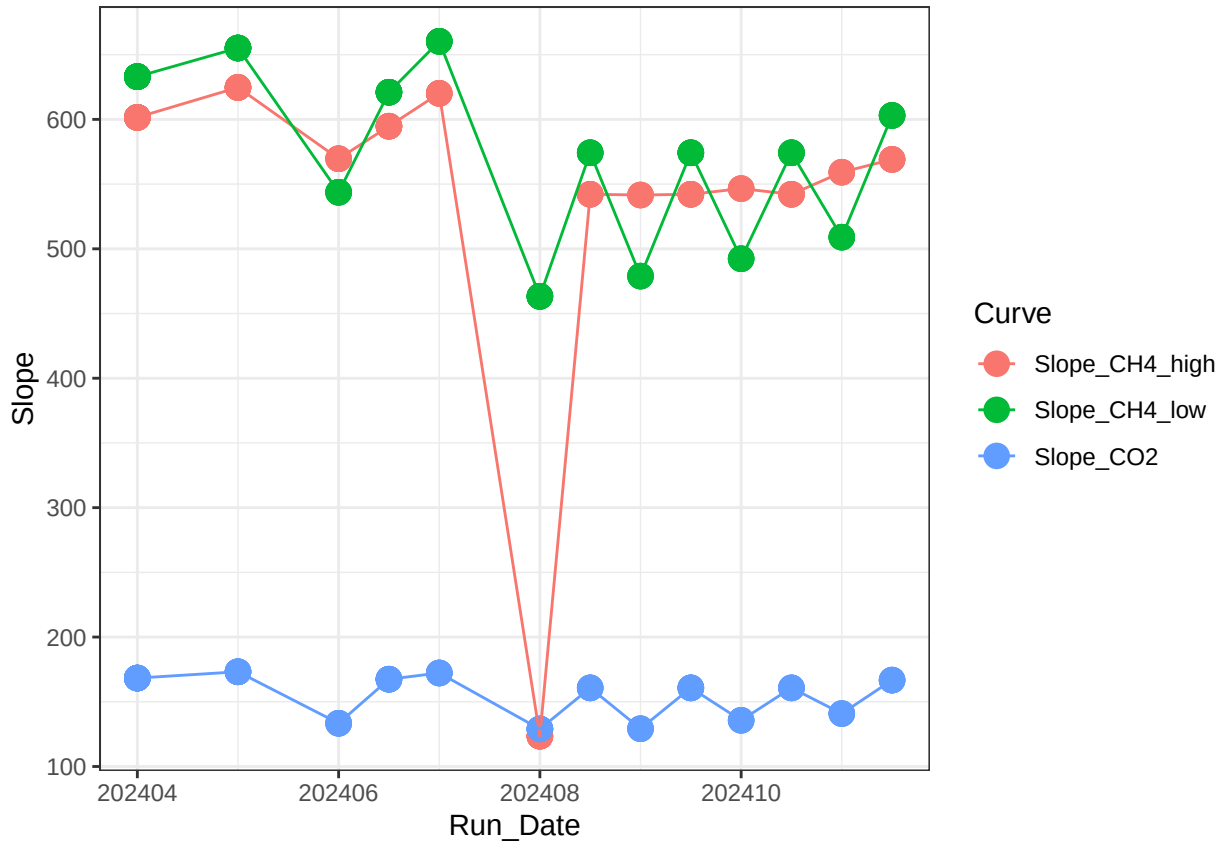

```
##
## Call:
## lm(formula = stds_ch4_low$CH4_Area ~ stds_ch4_low$STD_Conc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -616.47 -172.86  -94.26  265.08  476.33
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)             148.075    101.478   1.459    0.167
## stds_ch4_low$STD_Conc  620.984      2.493 249.109   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 326.5 on 14 degrees of freedom
## Multiple R-squared:  0.9998, Adjusted R-squared:  0.9998
## F-statistic: 6.206e+04 on 1 and 14 DF,  p-value: < 2.2e-16

## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## Call:
## lm(formula = stds_ch4$CH4_Area ~ stds_ch4$STD_Conc)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -78436 -13708 -12611  -5959 168718
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       13847.342  10713.934   1.292    0.209
## stds_ch4$STD_Conc   594.716      3.368 176.594   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 48900 on 24 degrees of freedom
## Multiple R-squared:  0.9992, Adjusted R-squared:  0.9992
## F-statistic: 3.119e+04 on 1 and 24 DF,  p-value: < 2.2e-16

## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## Call:
## lm(formula = stds_co2$CO2_Area ~ stds_co2$STD_Conc)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -75614  -8577  -5701   8199  74101
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       1.189e+04  1.160e+04   1.025     0.33
## stds_co2$STD_Conc 1.674e+02  5.646e-01 296.579   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 35420 on 10 degrees of freedom
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9999
## F-statistic: 8.796e+04 on 1 and 10 DF,  p-value: < 2.2e-16

##   X          Curve        R2    Slope  Intercept Run_Date
## 1 1  Slope_CH4_low 0.9996407 633.0314  -320.6965   202404
## 2 2 Slope_CH4_high 0.9993443 601.5512 14817.1191   202404
## 3 3      Slope_CO2 0.9999907 168.3200 10075.2183   202404
## 4 4  Slope_CH4_low 0.9996407 633.0314  -320.6965   202404
## 5 5 Slope_CH4_high 0.9993443 601.5512 14817.1191   202404
## 6 6      Slope_CO2 0.9999907 168.3200 10075.2183   202404
```

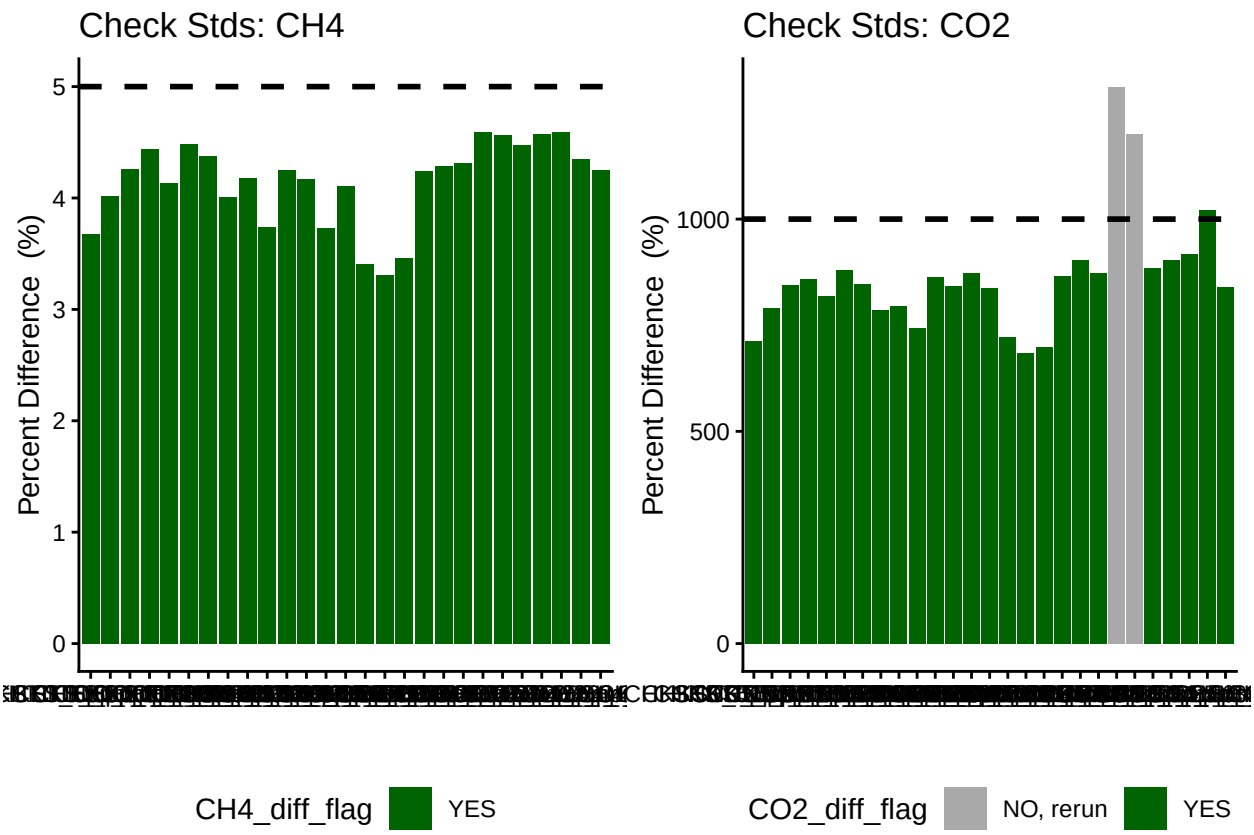


Now calculate the CH4 & CO2 concentrations in ppm

```
#head(raw)

#pull out methane standards
Samples <-  raw %>%
  filter(!str_detect(Sample_Type, "STD_CH4")) %>%
  filter(!str_detect(Sample_Type, "STD_CO2")) %>%
  filter(!str_detect(Sample_Type, "Blank")) %>%
  filter(!str_detect(Sample_Type, "Chk_STD")) %>%
   filter(!str_detect(Sample_Type, "CHKSTD")) %>%
  filter(!str_detect(Sample_Type, "NA"))
#head(Samples)

#Now flag any areas that are above the 100ppm area for CH4
Samples$CH4_Curve <- ifelse(Samples$CH4_Area >71000, "High", "Low")
#head(Samples)

#Calculate CH4 concentrations in ppm
Samples$CH4_Conc_ppm <- ifelse(Samples$CH4_Area >71000, (Samples$CH4_Area-Slope_CH4_high$Intercept)/Slo

#Calculate CO2 concentrations in ppm
Samples$CO2_Conc_ppm <- ((Samples$CO2_Area-Slope_CO2$Intercept)/Slope_CO2$Slope)
```

```
#head(Samples)

##########make flags for any dilutions needed
#highest CH4 standard = 10000
#highest CO2 standard = 50000

Samples$CH4_Flag <- ifelse(Samples$CH4_Conc_ppm >10000, "Needs Dilution", "Within Range")
Samples$CO2_Flag <- ifelse(Samples$CO2_Conc_ppm >50000, "Needs Dilution", "Within Range")
#head(Samples)
```

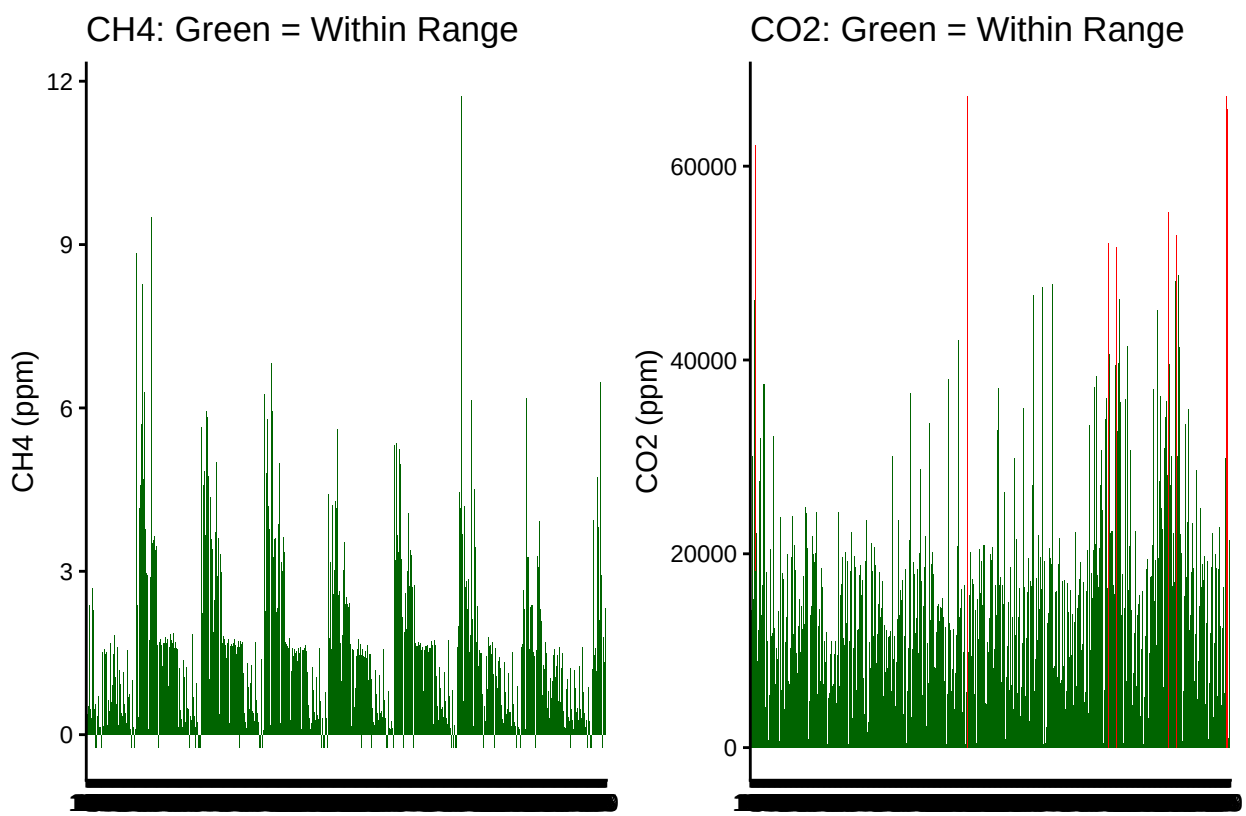
Check the Check Standards



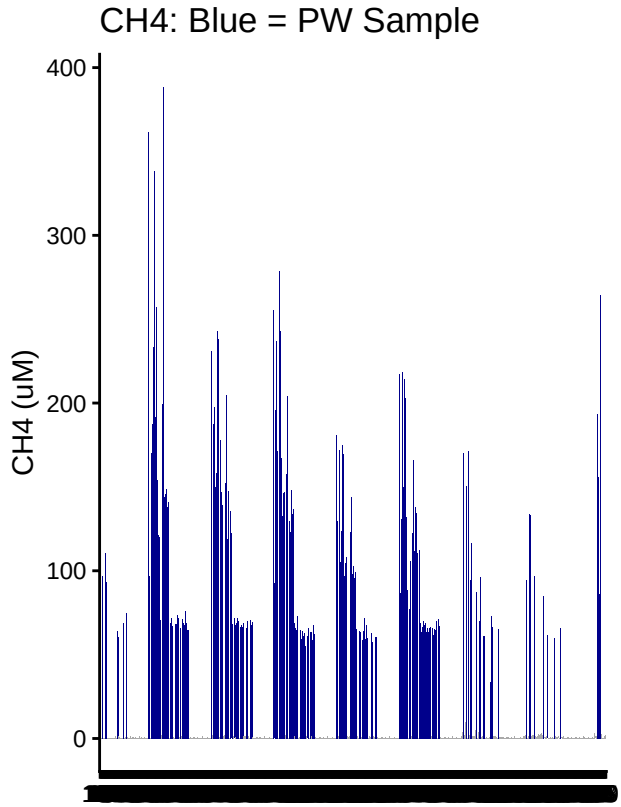
Dilution correct samples

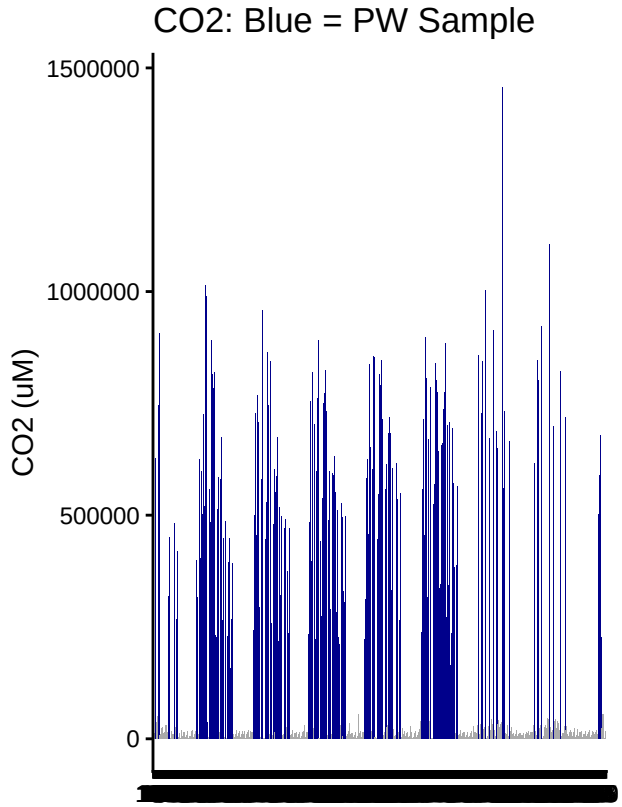
```
#multiply the concentration by the dilution factor
Samples$CH4_Conc_ppm_dilcorr <- (Samples$CH4_Conc_ppm * Samples$Dilution_Factor)

Samples$CO2_Conc_ppm_dilcorr <- (Samples$CO2_Conc_ppm * Samples$Dilution_Factor)

#check results
#head(Samples)
```

```r
#quick first look at the samples
ch4_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CH4_Conc_ppm, fill=CH4_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c("darkgreen","red"))+
        #scale_fill_gradient2(low='red', mid='white', high='blue', space='Lab') +
        theme_classic() + labs(x= " ", y="CH4 (ppm)", title="CH4: Green = Within Range") +
        theme(legend.position="none")


co2_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CO2_Conc_ppm, fill=CO2_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c("red", "darkgreen"))+
        #scale_fill_gradient2(low='red', mid='white', high='blue', space='Lab') +
        theme_classic() + labs(x= " ", y="CO2 (ppm)", title="CO2: Green = Within Range") +
        theme(legend.position="none")

ggarrange(ch4_samples, co2_samples, nrow=1, ncol=2)
```



If samples are water calculate gas in water





Write out processed data & slopes

```
#check results
#head(Samples)

#pull out what we need
Samples1 <- Samples[ ,c(1:3,6:17,21, 24:33)]
#head(Samples1)

#change the column names as needed
#colnames(samples1) <- c("Machine", "User", "Run_Date" , "Sample_Year" ,  "Sample_Month" , "Sample_Day"
 #                       "Sample_Time" , "Time_Zone" , "G_W" , "Sample_Plot","Sample_Grid","Sample_Dept
  #                      "Sample_ID" ,  "Dilution_Factor", "Notes" ,  "CH4_Flag", "CO2_Flag" ,
   #                     "CH4_Conc_ppm_dilcorr", "CO2_Conc_ppm_dilcorr")

write.csv(Samples1, "2024 Processed Data/COMPASS_TEMPEST_SGW_202406_Processed.csv")
```

#end